Block min-entropy dispersers yield sunflower and robust-sunflower compression

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- Finite set systems, block min-entropy sources and seeded extractors. Original definitions and parameter conventions retained with the complete construction.. No local mathematical proof omitted or independently certified.

1 Introduction

Let $\mathcal{F}$ be a family of sets from some universe X. A common theme and extensively studied phenomenon in combinatorics is the following: if the cardinality of $\mathcal{F}$ (when $\mathcal{F}$ is finite) or the density of $\mathcal{F}$ (when $\mathcal{F}$ is infinite) is large enough, then some nice patterns will occur in $\mathcal{F}$. Well-known examples of this kind include (1) Szemerédi's theorem [28], which asserts that all subsets of the natural numbers of positive density contain arbitrarily long arithmetic progressions; (2) Ramsey's Theorem [12], which asserts that if one colors the edges of a large enough complete graph with a finite number of colors, then there must exist a monochromatic clique of a certain size; and (3) the Erdős–Rado Sunflower Theorem [11], which asserts that a large enough family of sets of bounded size must contain a large sunflower.[1]

The study of these problems has resulted in many important tools (e.g., Szemerédi's Regularity Lemma [29] and the probabilistic method), which have found applications not only in combinatorics, or mathematics more broadly, but also in theoretical computer science (TCS). Conversely, ideas from TCS have influenced related research in combinatorics quite often. For example, the first two problems we mentioned above, Szemerédi's Theorem and Ramsey's Theorem, are intimately connected to the area of pseudorandomness in TCS. Indeed, by constructing a certain sparse pseudorandom subset of natural numbers and proving an appropriate Szemerédi-type theorem with respect to that subset, a celebrated result of Green and Tao [15] shows that prime numbers contain arbitrarily long arithmetic progressions. As for Ramsey's Theorem, a recent line of work on randomness extractors [3–5, 7, 17, 18] gives strongly explicit[2] constructions of Ramsey graphs that get close to the probabilistic bound [10].

[1] A sunflower is a family of sets whose pairwise intersection is constant, which we will formally define shortly.

[2] A class of graphs is strongly explicit if the vertices of each graph $G = (V, E)$ in the class are encoded by $(0, 1)$-strings of length $O(\log |V|)$ and a polynomial-time algorithm $P(|V|, i, j)$ decides adjacency of any pair (i, j) of vertices of G in polynomial (i. e., polylog($|V|$)) time.

In this paper we study the Sunflower Theorem and its related variants. We show that again there is an intimate connection to randomness extractors. In fact, using techniques from randomness extractors, we build a general proof framework that can unify the Sunflower Theorem and its variant known as the Robust Sunflower Lemma [27].

Sunflowers. A family is a set of sets. (Repeated sets are not permitted in counting the size of a family.) A family $\mathcal{F}$ is w-uniform if every element of $\mathcal{F}$ has size w. An r-sunflower is defined to be a family of r sets such that the intersection of any two sets in the family is the same set (which can be the empty set). Choose any w-uniform family $\mathcal{F}$, the main question of interest is, how large $\mathcal{F}$ needs to be in order to ensure that there is an r-sunflower in $\mathcal{F}$. Erdős and Rado proved the following theorem.

Theorem 1.1 (Sunflower Theorm, Erdős–Rado [11]). Let $\mathcal{F}$ be a w-uniform family. If $|\mathcal{F}| > w!(r-1)^w$ then $\mathcal{F}$ contains an r-sunflower.

They also conjectured that the bound on $|\mathcal{F}|$ can be replaced by c_r^w where c_r is a constant that only depends on r for every $r > 0$. This conjecture is one of the most well-known open problems in combinatorics, which remains open today despite extensive research.

The Sunflower Theorem has applications in TCS, such as in the proof of strong lower bounds for monotone circuits [26]. In addition, Alon, Shpilka and Umans [1] related the Sunflower Conjecture and its variants to possible approaches to fast matrix multiplication. Recently, following the breakthrough proof of the Cap-set Conjecture [8,9], a weaker version of the Sunflower Conjecture was also proved using the same method [24] (concretely, the Erdős–Szemerédi Sunflower Conjecture for $w = 3$). However, the general conjecture remains open.

Robust sunflowers. Motivated by the applications of sunflowers to proving monotone circuit lower bounds, robust sunflowers were introduced by Rossman [27] under the name "quasi-sunflowers" to prove monotone circuit lower bounds for the k-clique problem on random graphs. We denote by $\mathcal{P}(X)$ the family of all subsets of a finite set X.

Definition 1.2 (Robust sunflower [27]). Let X be a finite set and $\mathcal{S} \subseteq \mathcal{P}(X)$ a family of size $|\mathcal{S}| \geq 2$. Denote $Y = \bigcap_{T \in \mathcal{S}} T$. Following Rossman [27], for $p, \gamma \in [0,1]$, we say that $\mathcal{S}$ is a (p,γ)-robust sunflower if for a random set $W \subseteq X$, with each element of X present in W independently with probability p, it holds that

$$\Pr\left[\exists T \in \mathcal{S}, (T \setminus Y) \subseteq W\right] \geq 1 - \gamma.$$

As Rossman explains, the robust sunflower concept is a relaxation of the sunflower concept where petals may overlap in a limited way. Indeed, if $\mathcal{F}$ is a w-uniform sunflower then for every $p \in [0,1]$, $\mathcal{F}$ is a (p,γ)-robust sunflower with $\gamma = \exp(-|\mathcal{F}|p^w)$ [27, Remark 13].

In the same paper, Rossman also proved the following lemma, which says there is always a robust sunflower in a large family.

Lemma 1.3 (Robust Sunflower Lemma, Rossman [27]). Let $\mathcal{F}$ be a w-uniform family. If

$$|\mathcal{F}| \geq w! \cdot (1.71 \log(1/\gamma)/p)^w,$$

then[3] $\mathcal{F}$ contains a (p,γ)-robust sunflower.

Besides the original application, Rossman's Robust Sunflower Lemma was also used by Gopalan, Meka and Reingold [14] to study the problem of DNF sparsification. Given a DNF formula f on n variables, there are two natural ways to measure the complexity of f: the number of clauses (also called size) $s(f)$, and the maximum width of a clause $w(f)$. It is easy to show that any DNF of small size can be approximated well by another DNF of small width, by truncating clauses of large width. Gopalan et al. [14] used Rossman's Robust Sunflower Lemma to show the reverse direction, that any DNF with small width can also be approximated well by another DNF with small size. In particular, they showed that any width-w DNF formula can be ε-approximated by another DNF formula with size at most

[3] Throughout this paper, "log" refers to base-2 logarithms.

$(w \log(1/\varepsilon))^{O(w)}$. This kind of sparsification has applications in constructing pseudorandom generators and approximately counting the number of satisfying assignments for DNF formulas.

Similarly to the Sunflower Conjecture, one can also ask whether the bound on $|\mathcal{F}|$ in the Robust Sunflower Lemma can be improved. This question was further studied by Alweiss, Lovett, Wu and Zhang [2] and Rao [25]. We discuss further details in Section 1.4.

1.1 Our contribution

We provide a general framework to prove both the Sunflower Theorem and the Robust Sunflower Lemma. In fact, we reduce both of these problems to the construction of a certain type of randomness extractors. To state our results, we first formally define the notions that are going to be used in our extractors.

A random variable X, taking values in a finite set Σ, defines a probability measure on Σ, namely, for every subset $\Delta \subseteq \Sigma$ we have the measure $\mu(\Delta) = \Pr(X \in \Delta)$. Risking some confusion, we shall also refer to X as a "distribution over Σ," reducing the meaning of X to the probability measure it defines on Σ. We refer to Σ as the sample space. The support of X, denoted $\mathsf{Supp}(X)$, is the smallest subset $\Delta \subseteq \Sigma$ such that $\Pr(X \in \Delta) = 1$.

Definition 1.4. Let X be a distribution over a finite sample space Σ. The min-entropy of X is defined as

$$\mathcal{H}_\infty(X) = \min_{x \in \Sigma} \left\{ \log \left(\frac{1}{\Pr[X = x]} \right) \right\}.$$

A distribution over $\{0,1\}^n$ is said to be an (n,k) source if the distribution has min-entropy at least k.

Definition 1.5 (Block min-entropy source). Let $X_1, \ldots, X_m$ be random variables with sample space $\{0,1\}^n$. Consider the random variable $X = (X_1, \ldots, X_m)$ (with sample space $\{0,1\}^{nm}$). The distribution X is an (m,n,k) block min-entropy source if for every non-empty subset $S \subseteq [m]$, the joint distribution of $(X_i : i \in S)$ has min-entropy at least $k|S|$.

We note that the definition of block min-entropy sources was initiated in [13] as a tool to prove lifting theorems in communication complexity.

Definition 1.6 (Block min-entropy extractor). A function $E : \{0,1\}^n \times \{0,1\}^s \to \{0,1\}^d$ is a (k, ε, d, s) block min-entropy extractor if for any $m, n \in \mathbb{N}$ and any (m,n,k) block min-entropy source $X = (X_1, \ldots, X_m)$, it holds that

$$(E(X_1, R_1), \ldots, E(X_m, R_m)) \approx_\varepsilon U_{dm}.$$

Here, each $R_i \in \{0,1\}^s$ is an independent uniform random string, U_{dm} is the uniform distribution over (dm)-bit strings, and $\approx_\varepsilon$ means ε-close in statistical distance. If in addition it holds that

$$(E(X_1, R_1), R_1, \ldots, E(X_m, R_m), R_m) \approx_\varepsilon U_{(d+s)m},$$

then we say that the function E is a strong (k, ε, d, s) block min-entropy extractor.

We also define a weaker object called a disperser.

Definition 1.7 (Block min-entropy disperser). A function

$$E : \{0,1\}^n \times \{0,1\}^s \to \{0,1\}^d$$

is a (k, ε, d, s) block min-entropy disperser if for any $m, n \in \mathbb{N}$ and any (m,n,k) block min-entropy source $X = (X_1, \ldots, X_m)$, it holds that

$$|\,\mathsf{Supp}(E(X_1, R_1), \ldots, E(X_m, R_m))| \geq (1 - \varepsilon) 2^{dm}.$$

Here, each $R_i \in \{0,1\}^s$ is an independent uniform random string. If in addition there exists at least one way to fix $R_1 = r_1, \ldots, R_m = r_m$ such that

$$|\,\mathsf{Supp}(E(X_1, r_1), \ldots, E(X_m, r_m))| \geq (1 - \varepsilon) 2^{dm},$$

then we say that the function E is a strong (k, ε, d, s) block min-entropy disperser.

In this paper, we make connections between the block min-entropy disperser and sunflower and robust sunflower structures. Formally, we prove the following result.

Theorem 1.8. Suppose that there exists a strong $(k, 0, d, s)$ block min-entropy disperser, $E : \{0,1\}^n \times \{0,1\}^s \to \{0,1\}^d$ for any (w, n, k) block min-entropy source. Then the following holds.

Let $\mathcal{F}$ be a w-uniform family. Assume that $|\mathcal{F}| \geq 2^{(k+2)w}$. Then:

(i) $\mathcal{F}$ contains a 2^d-sunflower.

(ii) $\mathcal{F}$ contains a $\left(p, w(1-p)^{2^d}\right)$-robust sunflower.

Observe that the seed length s of the extractor does not play a part in the conclusion of Theorem 1.8. We then show that we can construct strong zero-error block min-entropy extractors. As a corollary, we are able to construct strong block min-entropy dispersers. Specifically, we have the following result.

Theorem 1.9. There is a constant $c > 1$ such that for any $m, n, k \in \mathbb{N}$ with $k \geq c \log m$, we have:

- There is an explicit[4] strong (k, ε, d, s) block min-entropy extractor $E : \{0,1\}^n \times \{0,1\}^s \to \{0,1\}^d$ for (m, n, k) block min-entropy sources, where $s = n$, $d = k/c$ and $\varepsilon = 2^{-\Omega(k)}$.
- There is an explicit strong $(k, 0, d, s)$ block min-entropy disperser $E : \{0,1\}^n \times \{0,1\}^s \to \{0,1\}^d$ for (m, n, k) block min-entropy sources, where $s = n$, $d = k/c$.

Combined with Theorem 1.8, this gives the Sunflower Theorem and the Robust Sunflower Lemma. In Theorem 1.9, we give explicit constructions. We also note that the existence of block min-entropy dispersers is also enough to prove sunflower lemmas.

Corollary 1.10 (Sunflower Theorem, this paper). There is a constant c such that for any w-uniform family and any $r > 1$, if $|\mathcal{F}| \geq (wr)^{cw}$, then $\mathcal{F}$ contains an r-sunflower.

The reader will note that this result is weaker than Theorem 1.1, the original result by Erdős and Rado. The point we are making here is that our result is comparable, and it is obtained using a different approach that may lead to improved bounds. Indeed, a variant of our method did lead to improved bounds [2, 25].

Corollary 1.11 (Robust Sunflower Lemma, this paper). There is a constant c such that for any w-uniform family $\mathcal{F}$, if $|\mathcal{F}| \geq \left(\frac{w+\log(1/\gamma)}{p}\right)^{cw}$, then $\mathcal{F}$ contains a (p, γ)-robust sunflower.

1.2 Overview of the techniques

Our reduction from sunflower/robust sunflower problems to block min-entropy dispersers is as follows. Suppose the family $\mathcal{F} \subseteq \mathcal{P}(X)$ for some set X, where each set in $\mathcal{F}$ has size w. We first show that without loss of generality we can assume $\mathcal{F}$ is a w-partite family.

Definition 1.12 (w-partite family). Let X be a finite set and let $\mathcal{F} = \{U_i\}_{i \in I}$ be a family of subsets of X. We say that $\mathcal{F}$ is a w-partite family if

- $\mathcal{F}$ is w-uniform;
- there is a partition $X_1, \ldots, X_w$ of X such that for every $U \in \mathcal{F}$, we have $|X_j \cap U| = 1$ for each $j \in [w]$.

Consider the uniform distribution over a w-partite family $\mathcal{F}$. There are two possible cases:

- Case 1: there is a subset S which is a subset of many elements of $\mathcal{F}$, specifically,

$$\left|\{U \in \mathcal{F} : S \subseteq U\}\right| \geq \frac{|\mathcal{F}|}{r^{|S|}},$$

where r is a parameter to be determined.

[4] A class of finite functions is explicit if a polynomial-time algorithm evaluates each function in the class on each of its inputs. The class of bipartite graphs corresponding to an explicit class of functions is strongly explicit in the sense of Footnote 2.

- Case 2: every set S does not appear in too many sets of $\mathcal{F}$.

In Case 1, S is already like a core in a sunflower or robust sunflower, thus we can apply induction on the subfamily $\mathcal{F}_S := \{U \setminus S : S \subseteq U \in \mathcal{F}\}$. In Case 2, the condition basically implies that the distribution is relatively flat, which equivalently translates into a block min-entropy source as we defined above. One can naturally imagine that the worst-case situation here is that the distribution is actually the uniform distribution over $X_1 \times \cdots \times X_w$, and we show that indeed this is the case by using our zero-error block min-entropy disperser (Theorem 1.9). It is then easy to see that in the worst case, the empty set is a robust sunflower, or one can choose a sunflower with size 2^d (the support size in the output of the disperser) whose core is the empty set.

1.3 The role of extractors in our reduction

One can view the block min-entropy extractor/disperser used in our reduction as a gadget, which reduces the sunflower/robust sunflower problem in the general case to the much easier case of a uniform distribution (or full support) on $X_1 \times \cdots \times X_w$. This is similar to the role of extractors in recent work that showed lifting theorems from query complexity to communication complexity [13], and linear programming lower bounds for constraint satisfaction problems [16].

In fact, the extractors used in these articles are essentially the same as the extractors used in the present paper (although here we need to show that the extractor/disperser is strong, while in [13] and [16] this is not necessary), and the barriers to further improvement are also similar. Specifically, in all such constructions one needs the min-entropy $k \geq c \log m$ for some constant $c > 1$, where m is equal to the size of the sets (i.e., w) in our applications. It is interesting to ask whether $k = \Omega(\log m)$ is necessary. Any improvement of such extractors leads to improvements of both sunflower theorems and lifting theorems. Subsequent to the conference version of this paper [19], Meka [23] gave a counterexample showing that such extractors do not exist.

1.4 Subsequent work

In this section we report work done after the publication of the conference version of this paper [19].

Connection to DNF sparsification: The connection of sunflowers and DNF sparsification was first discovered by Gopalan, Meka and Reingold [14]. Building on the Robust Sunflower Lemma, Gopalan et al. [14] proved any width-w DNF (with arbitrary size) can be ε-approximated by a DNF of size at most $(w \cdot \log(1/\epsilon))^{O(w)}$. Similarly to the Sunflower Conjecture, Gopalan et al. also believed the DNF compression bound can be improved to $(\log(1/\epsilon))^{O(w)}$. This conjecture was confirmed by subsequent work by Lovett, Wu and Zhang [21, 22].

In another piece of subsequent work Lovett, Solomon and Zhang [20] also proved that the reverse direction is also true. That is, an improved upper bound on DNF sparsification implies an improved sunflower bound.

Progress on the Sunflower Conjecture. In this paper, we show that any w-uniform family $\mathcal{F}$ of size $|\mathcal{F}| \geq w^{cw}$ for some constant $c > 1$ contains a 3-sunflower. Furthermore, we show that any family $\mathcal{F}$ with the following spread condition must contain 3 pairwise disjoint sets.

Definition 1.13. Given a family $\mathcal{F}$ and $r > 0$, we say that $\mathcal{F}$ is r-spread, if for any set S,

$$\big|\{U \in \mathcal{F} : S \subseteq U\}\big| \leq \frac{|\mathcal{F}|}{r^{|S|}}.$$

The following corollary follows directly from the proof of Theorem 1.8.

Corollary 1.14. There is a constant $c > 0$, such that for any w-partite family $\mathcal{F}$, if $\mathcal{F}$ is w^c-spread then it contains w pairwise disjoint sets.

This w^c-spread condition actually comes from the requirement of our disperser that $k \geq c \log w$. As discussed above, it is interesting to ask whether this is necessary. In an article subsequent to the conference version of this paper [19], Alweiss, Lovett, Wu and Zhang [2] improved the w^c bound on the spread to $(\log w)^{O(1)}$ and Rao [25] further improved it to $O(\log w)$.

Lemma 1.15 ([2, 25]). There is a constant $c > 0$, such that for any w-partite family $\mathcal{F}$, if $\mathcal{F}$ is $(cr \cdot \log(wr))$-spread then it contains r pairwise disjoint sets.

This lemma finally leads to an improved sunflower lemma.

Corollary 1.16 ([2, 25]). There is a constant $c > 0$ such that, for each w-uniform family $\mathcal{F}$, if $|\mathcal{F}| > (cr \cdot \log(wr))^w$ then $\mathcal{F}$ contains an r-sunflower.

Discussion of robust sunflowers. In this paper, we also study robust sunflower structures. In particular, we have the following corollary. Below, we use the notation $O_{p,\gamma}(\cdot)$ to hide the specific dependency on the parameters p, γ, which is of less interest to us here.

Corollary 1.17. There is a constant $c > 0$ such that, for any w-partite family $\mathcal{F}$, if $\mathcal{F}$ is r-spread where $r = (O_{p,\gamma}(w))^c$, then $\mathcal{F}$ is a (p, γ)-robust sunflower with empty kernel.

The subsequent work of Alweiss, Lovett, Wu and Zhang [2], and its improvement by Rao [25], gives an improved robust sunflower lemma.

Lemma 1.18 ([2], [25]). There is a constant $c > 0$ such that, for any w-partite family $\mathcal{F}$, if $\mathcal{F}$ is $(c \log(w/\gamma)/p)$-spread then $\mathcal{F}$ is a (p, γ)-robust sunflower with empty kernel.

Furthermore, the $\log w$ term is tight by the following example. Fix $p = \gamma = 1/2$ for convenience. Let $X_1, \ldots, X_w$ be w disjoint sets each of size $c \log w$ for some large enough $c > 0$. Define the family $\mathcal{F} := X_1 \times \cdots \times X_w$. By this we mean, with some abuse of notation, the complete w-partite hypergraph on $\{X_1, \ldots, X_w\}$. Then $\mathcal{F}$ does not contain a (p, γ)-robust sunflower. This example also sets a barrier to proving the Sunflower Conjecture via robust sunflowers.

Discussion of the block min-entropy extractor and the disperser construction. In this paper, we construct explicit strong block min-entropy extractors and dispersers. However, in our constructions, we require the min-entropy k to be as large as $\Omega(\log m)$. In the conference version of this paper we asked whether the min-entropy condition could be improved to be $o(\log m)$. The following counterexample due to Raghu Meka [23] shows that this is impossible. Therefore it sets a barrier to improving (robust) sunflowers via extractors.

Theorem 1.19. Let m, n be parameters and set $k := \min(\log(2^n - 1), \log m)$. Then for any function $E : \{0,1\}^n \times \{0,1\}^s \to \{0,1\}$, there is an (m, n, k) block min-entropy source $X = (X_1, \ldots, X_m)$ such that, for every $r_1, \ldots r_m \in \{0,1\}^s$, the support of

$$(E(X_1, r_1), \ldots, E(X_m, r_m))$$

is not full.

Proof. Let $E : \{0,1\}^n \times \{0,1\}^s \to \{0,1\}$ be the given function. Fix an arbitrary string $x^* \in \{0,1\}^n$. Define the distribution $X = (X_1, \ldots, X_m)$ as follows. First, sample an index $i^* \in [m]$ uniformly and set $X_{i^*} = x^*$. For every $i \neq i^*$, sample $X_i \in \{0,1\}^n \setminus \{x^*\}$ uniformly and independently. Observe that X is a (m, n, k) block min-entropy source for $k = \min(\log(2^n - 1), \log m)$.

We will show that for every $r_1, \ldots, r_m \in \{0,1\}^s$, the random variable $(E(X_1, r_1), \ldots, E(X_m, r_m))$ does not have full support on $\{0,1\}^m$. This is because if we let $z = (E(x^*, r_1), E(x^*, r_2), \ldots, E(x^*, r_m))$, then the output of $(E(X_1, r_1), \ldots, E(X_m, r_m))$ always agrees with z in at least one position. Thus the string $z \oplus 1$ is not in the support of $(E(X_1, r_1), \ldots, E(X_m, r_m))$.

□

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2 Preliminaries

We first review some basic definitions in probability.

Definition 2.1. Let D be a distribution over a finite sample space Σ. Its entropy is

$$\mathcal{H}(D) = \sum_{x \in \Sigma} \Pr[D = x] \cdot \log\left(\frac{1}{\Pr[D = x]}\right).$$

Its min-entropy is

$$\mathcal{H}_\infty(D) = \min_{x \in \Sigma} \left\{ \log\left(\frac{1}{\Pr[D = x]}\right) \right\}.$$

Its max-entropy is

$$\mathcal{H}_0(D) = \log |\,\mathsf{Supp}(D)|.$$

Definition 2.2 (Statistical distance). Let D_0 and D_1 be distributions over a finite sample space Σ. The statistical distance between D_0 and D_1 is defined as

$$\mathrm{dist}(D_0, D_1) = \frac{1}{2} \sum_{x \in \Sigma} \big|\Pr[D_0 = x] - \Pr[D_1 = x]\big|.$$

3 A construction of a block min-entropy extractor

We use the following well-known extractor based on the inner product function [6]. We denote by $\mathbb{F}_q$ the finite field on q elements. When $q = 2^\ell$ we identify $\mathbb{F}_q$ with $\{0,1\}^\ell$ and $\mathbb{F}_q^t$ with $\{0,1\}^{t\ell}$.

Theorem 3.1. Let $t, \ell \geq 1$ and take $q = 2^\ell, n = t\ell$. Let X, Y be independent sources on $\mathbb{F}_q^t \cong \{0,1\}^n$ with min-entropies k_1, k_2, respectively. Let IP be the inner product function over the field $\mathbb{F}_q$. Then:

$$\mathrm{dist}\left((\mathrm{IP}(X,Y), X), (U_\ell, X)\right) \leq \varepsilon \quad \text{and} \quad \mathrm{dist}\left((\mathrm{IP}(X,Y), Y), (U_\ell, Y)\right) \leq \varepsilon$$

where $\varepsilon = 2^{-(k_1 + k_2 - n - \ell)/2}$.

Now we can construct a block min-entropy extractor as follows. Given parameters n, k, choose a field $\mathbb{F}_q$ such that $q = 2^\ell$ with $\ell = \alpha k$ for some constant $0 < \alpha < 1$ to be determined later. Without loss of generality we assume that $n = t\ell$ for some integer t. We view $X \in \{0,1\}^n$ as a vector in $\mathbb{F}_q^t$ and choose a uniform independent seed $R \in \{0,1\}^n \cong \mathbb{F}_q^t$.

A block min-entropy extractor

1. Given parameters m, n, k let q, t be as described above.
2. Sample $(X_1, \ldots, X_m) \in (\mathbb{F}_q^t)^m$ from the block min-entropy distribution.
3. Sample $(R_1, \ldots, R_m) \in (\mathbb{F}_q^t)^m$ uniformly and independently.
4. Output $Z := (\mathrm{IP}(X_1, R_1), \ldots, \mathrm{IP}(X_m, R_m))$.

We are now ready to prove the following theorem.

Theorem 3.2 (Theorem 1.9 restated). Let $X = (X_1, \ldots, X_m)$ be an (m, n, k) block min-entropy source. Let $Z \in \{0,1\}^{\ell m}$ be the output of the above block min-entropy extractor applied to X. There exists a constant $c > 1$ such that if $k \geq c \log m$, then the following holds for error $\varepsilon = 2^{-\Omega(k)}$:

- With probability $1 - \varepsilon$ over the fixing of the seed $(R_1, \ldots, R_m)$,

$$\left|\Pr[Z = z] - 2^{-\ell m}\right| \leq \varepsilon \cdot 2^{-\ell m} \qquad \forall z \in \{0,1\}^{\ell m}.$$

  In particular, in such cases $\mathcal{H}_0(Z) = \ell m$

- $\mathrm{dist}\left((Z, R_1, \ldots, R_m), (U, R_1, \ldots, R_m)\right) \leq 2\varepsilon$.

Proof. We have a joint distribution $(X_1, \ldots, X_m)$ that has block min-entropy k. The output of the local extractor applied to $(X_1, \ldots, X_m)$, using m independent uniform seeds $(R_1, \ldots, R_m)$, is a distribution $(Z_1, \ldots, Z_m)$ over $\{0,1\}^{\ell m} = \mathbb{F}_q^m$ where $Z_i = \mathrm{IP}(X_i, R_i)$ for each i.

For any fixing of the seed $(R_1 = r_1, \ldots, R_m = r_m)$, the distribution $(Z_1, \ldots, Z_m)$ is a deterministic function of $(X_1, \ldots, X_m)$, and we will view this distribution as a function $\mathcal{D} : \{0,1\}^{\ell m} \to [0,1]$ where the image of each input is its associated probability in the distribution. We now write this function in its Fourier basis:

$$\mathcal{D}(z) = \sum_{S \subseteq [\ell m]} \hat{\mathcal{D}}(S) \chi_S(z),$$

where $z = (z_1, \ldots, z_m) \in \{0,1\}^{\ell m}$, $\chi_S(z) = (-1)^{\sum_{i \in S} z(i)} \in \{+1, -1\}$, and

$$\hat{\mathcal{D}}(S) = 2^{-\ell m} \cdot \sum_z \mathcal{D}(z) \chi_S(z) = 2^{-\ell m} \cdot \mathbb{E}_{z \sim \mathcal{D}}[\chi_S(z)].$$

Here we use $z(i)$ to stand for the i-th bit of the string z. This is to distinguish between the notation z_i, which refers to the i-th block of the string z, that contains ℓ bits.

Note that $\hat{\mathcal{D}}(\emptyset) = 2^{-\ell m}$ since $\mathcal{D}$ is a probability distribution. Thus we have that $\forall z \in \{0,1\}^{\ell m}$,

$$\left|\mathcal{D}(z) - 2^{-\ell m}\right| = \left|\sum_{S \subseteq [\ell m], S \neq \emptyset} \hat{\mathcal{D}}(S) \chi_S(z)\right| \leq \sum_{S \subseteq [\ell m], S \neq \emptyset} \left|\hat{\mathcal{D}}(S)\right|. \tag{3.1}$$

Note that for any $S \subseteq [\ell m]$, $\chi_S(Z)$ corresponds to the parity of a subset of the bits in Z. For each $Z_j, j \in [m]$, this parity may or may not involve any bits in Z_j. We will be interested in the number of indices j such that $\chi_S(Z)$ involves at least one bit from Z_j, and we call this number $\Delta(S)$. Note that $\Delta(\emptyset) = 0$ and $1 \leq \Delta(S) \leq m$ for any $S \neq \emptyset$.

We now have the following lemma.

Lemma 3.3. If $\Delta(S) = h$, then with probability $1 - 2^{-h(1-\alpha)k/4}$ over the fixing of the seed $(R_1 = r_1, \ldots, R_m = r_m)$, we have that $|\hat{\mathcal{D}}(S)| \leq 2 \cdot 2^{-\ell m} 2^{-h(1-\alpha)k/4}$.

Proof. Without loss of generality assume that the Z_j from which $\chi_S(Z)$ involves at least one bit are $(Z_1, \ldots, Z_h)$. Note that for any $Z_i \in \{0,1\}^\ell = \mathbb{F}_q$, any parity of the bits of Z_i corresponds exactly to the first bit of $a \cdot Z_i$ viewed as a vector in $\{0,1\}^\ell$, for some $a \in \mathbb{F}_q$ and where the operation $\cdot$ is multiplication in the field $\mathbb{F}_q$. Moreover this correspondence is a bijection in the sense that different parities correspond to different elements $a \in \mathbb{F}_q$. The special case of parity over the empty set corresponds to the case of $a = 0$. Thus, $\sum_{i \in S} Z(i)$ corresponds to the first bit of $\sum_{j \in [h]} a_j Z_j$ viewed as a vector in $\{0,1\}^\ell$, for some non-zero $\{a_j \in \mathbb{F}_q : j \in [h]\}$. Note that

$$\sum_{j \in [h]} a_j Z_j = \sum_{j \in [h]} a_j \mathrm{IP}(X_j, R_j) = \sum_{j \in [h]} \mathrm{IP}(a_j X_j, R_j) = \mathrm{IP}((a_1 X_1, \ldots, a_h X_h), (R_1, \ldots, R_h)). \tag{3.2}$$

Since each $a_j \neq 0$ the transformation from $(x_1, \ldots, x_h)$ to $(a_1 x_1, \ldots, a_h x_h)$ is a bijection. Thus we know the distribution $(a_1 X_1, \ldots, a_h X_h)$ has min-entropy kh, while $(R_1, \ldots, R_h)$ has min-entropy nh. Thus by Theorem 3.1 applied over the field $\mathbb{F}_{2^{\ell h}}$ we have that

$$\mathrm{dist}\left(\left(\sum_{j\in[h]} a_j Z_j, R_1, \ldots, R_m\right), (U_{\ell h}, R_1, \ldots, R_m)\right) \leq 2^{\frac{-(h(k-\ell))}{2}} = 2^{\frac{-h(1-\alpha)k}{2}}. \tag{3.3}$$

In particular, as $\chi_S(Z)$ is the first bit of $\sum_{j\in[h]} a_j Z_j$, we have

$$\mathrm{dist}\left(\chi_S(Z), R_1, \ldots, R_m\right), (U_1, R_1, \ldots, R_m)) \leq 2^{\frac{-h(1-\alpha)k}{2}}. \tag{3.4}$$

By Markov's inequality this means that with probability $1 - 2^{-h(1-\alpha)k/4}$ over the fixing of the seed $R = (R_1, \ldots, R_m)$, we have $|\hat{\mathcal{D}}(S)| = |2^{-\ell m} \cdot \mathbb{E}_{z\sim\mathcal{D}}[\chi_S(z)]| \leq 2 \cdot 2^{-\ell m} 2^{-h(1-\alpha)k/4}$. □

Next, note that the number of S with $\Delta(S) = h$ is $\binom{m}{h}(2^\ell - 1)^h \leq 2^{(\ell+\log m)h}$. Recall that $\ell = \alpha k$ and $k \geq c \log m$. We can choose the constants α, c such that $2^{\ell+\log m} 2^{-(1-\alpha)k/4} \leq 2^{-k/8}$. Now we have as long as $k \geq 8$,

$$\sum_{h=1}^{m} \binom{m}{h} (2^\ell - 1)^h 2^{\frac{-h(1-\alpha)k}{4}} \leq \sum_{h=1}^{m} 2^{-\frac{hk}{8}} \leq 2^{-\frac{k}{8}+1}. \tag{3.5}$$

Set $\varepsilon = 2^{-k/8+2} = 2^{-\Omega(k)}$.

By the union bound we have that with probability at least $1 - \varepsilon$ over the fixing of the seed $(R_1 = r_1, \ldots, R_m = r_m)$, for every $S \neq \emptyset$ with $\Delta(S) = h$, $|\hat{\mathcal{D}}(S)| \leq 2 \cdot 2^{-\ell m} 2^{\frac{-h(1-\alpha)k}{4}}$. Thus for any such seed we have that

$$\left|\mathcal{D}(z) - 2^{-\ell m}\right| \leq \sum_{S\subseteq[\ell m], S\neq\emptyset} \left|\hat{\mathcal{D}}(S)\right| \leq \varepsilon \cdot 2^{-\ell m}. \tag{3.6}$$

This concludes the proof of the first part of Theorem 1.9. For the second part, notice that conditioned on the fixing of any seed $R_1, \ldots, R_m$, with probability $1 - \varepsilon$ the statistical distance is at most ε, and otherwise it is trivially bounded by 1. So overall the statistical distance between $(Z, R_1, \ldots, R_m)$ and $(U_{\ell m}, R_1, \ldots, R_m)$ is at most 2ε. □

4 Compressing set systems by the block min-entropy extractor

In this section, we focus on the set systems that satisfy the spread condition, and show a compression operator for such set systems. Our compression is based on the block min-entropy extractor. We first show that it suffices to consider w-partite families (see Definition 1.12).

Lemma 4.1. Let $\mathcal{F}$ be a w-uniform family. Then $\mathcal{F}$ has a w-partite subfamily $\mathcal{F}'$ of size $|\mathcal{F}'| \geq |\mathcal{F}|/2^{2w}$.

Proof. Let $U \in \mathcal{F}$ be a set, and let $X_1, \ldots, X_w$ be a random partition of X. Then

$$\Pr_{X_1,\ldots,X_w}[\forall j \in [w], |U \cap X_j| = 1] = \frac{w!}{w^w}.$$

Then, by averaging, there is a partition $(X_1, \ldots, X_w)$ such that

$$|\{U \in \mathcal{F} : \forall j \in [w], |U \cap X_j| = 1\}| \geq |\mathcal{F}| \cdot \frac{w!}{w^w}$$

The claim then follows since $\frac{w!}{w^w} \geq 2^{-2w}$. □

Now we can focus on w-partite families. Given a finite set X, we denote by X_p the distribution over subsets $W \subset X$, where each $x \in X$ appears in W independently with probability p.

Lemma 4.2. Let $u \geq w$. Let c be the constant from Theorem 1.9. Then for every w-partite family which is u^c-spread (recall Definition 1.13), it holds that

$$\Pr_{W\sim X_p}[\exists U \in \mathcal{F}, U \subseteq W] \geq 1 - w(1-p)^u.$$

To prove this lemma, we first define a "worst case" instance, and then show that all other instances behave better than this case. Let $X_1^*, \ldots, X_w^*$ be w disjoint sets each of size u. Define the family $\mathcal{U}^*$ as

$$\mathcal{U}^* = \left\{ \{x_1, \ldots, x_w\} : \forall j \in [w], x_j \in X_j^* \right\}.$$

Claim 4.3. Let $\mathcal{U}^*$ as defined above. Then

$$\Pr_{W \sim X_p}[\exists U \in \mathcal{U}^*, U \subseteq W] \geq 1 - w(1-p)^u.$$

Proof. By the definition of $\mathcal{U}^*$, we have that

$$\begin{aligned}
\Pr_W[\forall U \in \mathcal{U}^*, U \not\subseteq W] =& \Pr_W[\exists j \in [w], X_j \cap W = \emptyset] \\
\leq& \sum_{j \in [w]} \Pr_W[X_j \cap W = \emptyset] \\
=& w(1-p)^u.
\end{aligned}$$

□

Let X, Y be finite sets, $h : X \to Y$ a map. Given a set $U \subset X$ define $h(U) = \{h(x) : x \in U\} \subset Y$. Given a family $\mathcal{F} \subseteq \mathcal{P}(X)$ define $h(\mathcal{F}) \subseteq \mathcal{P}(Y)$ as

$$h(\mathcal{F}) = \{h(U) : U \in \mathcal{F} \text{ and } h \text{ is injective on } U\}.$$

Lemma 4.4. Let X and Y be sets, $h : X \to Y$ a map, $\mathcal{F} \subset \mathcal{P}(X)$. Then

$$\Pr_{W_Y \sim Y_p}[\exists U \in h(\mathcal{F}), U \subseteq W_Y] \leq \Pr_{W_X \sim X_p}[\exists U \in \mathcal{F}, U \subseteq W_X].$$

Proof. Without loss of generality, we can assume the map h is surjective, because elements $y \in Y \setminus h(X)$ do not affect the events. If $|Y| = |X|$ then h is a bijection and hence $\mathcal{F}$ and $h(\mathcal{F})$ are the same, up to renaming the elements. So, assume $|Y| < |X|$. It suffices to prove the lemma for the case that $|Y| = |X| - 1$, as the general case follows from applying this case iteratively (namely, decompose h as a sequence of maps, each reduces the domain size by one).

So, assume $|Y| = |X| - 1$. In this case, there is a unique pair $x_1, x_2 \in X$ such that $h(x_1) = h(x_2) = y$. We may assume without loss of generality (by renaming the elements of Y) that h is the identity map on $X' = X \setminus \{x_1, x_2\}$. This allows us to jointly sample (W_X, W_Y) as follows. Sample $W' \sim X'_p, W'_X \sim \{x_1, x_2\}_p, W'_Y \sim \{y\}_p$ and set $W_X = W' \cup W'_X, W_Y = W' \cup W'_Y$. We will show that for every fixed $W' = w'$,

$$\Pr_{W_Y \sim Y_p}[\exists U \in h(\mathcal{F}), U \subseteq W_Y \mid W' = w'] \leq \Pr_{W_X \sim X_p}[\exists U \in \mathcal{F}, U \subseteq W_X \mid W' = w']. \tag{4.1}$$

The lemma then follows by averaging over W'.

To that end, fix W'. Let $\mathcal{F}' = \{U \setminus X' : U \in \mathcal{F}, (U \cap X') \subset W'\}$. Note that $\mathcal{F}' \subseteq \mathcal{P}(\{x_1, x_2\})$. Similarly, define $\mathcal{F}'' = \{U \setminus X' : U \in h(\mathcal{F}), (U \cap X') \subset W'\}$. Note that $\mathcal{F}'' \subseteq \mathcal{P}(\{y\})$. Equation (4.1) is equivalent to

$$\Pr_{W'_Y \sim \{y\}_p}[\exists U \in \mathcal{F}'', U \subseteq W'_Y] \leq \Pr_{W'_X \sim \{x_1, x_2\}_p}[\exists U \in \mathcal{F}', U \subseteq W'_X]. \tag{4.2}$$

We verify Equation (4.2) by a case analysis.

(i) If $\mathcal{F}''$ is empty then the LHS of Equation (4.2) is 0, while the RHS is non-negative.

(ii) If $\emptyset \in \mathcal{F}''$ then $\emptyset \in \mathcal{F}'$. In this case, both the LHS and RHS of Equation (4.2) equal 1.

(iii) If $\mathcal{F}'' = \{\{y\}\}$ then either $\{x_1\} \in \mathcal{F}'$ or $\{x_2\} \in \mathcal{F}'$ (or possibly both). In either case, the LHS of Equation (4.2) equals p, while the RHS is at least p.

□

We now prove Lemma 4.2. Let $\mathcal{F}$ be a family that satisfies the assumptions. We will show there is a function h such that $h(\mathcal{F}) = \mathcal{U}^*$. The extractor from Theorem 1.9, with an appropriate choice of seed, provides such a function h.

Proof of Lemma 4.2. Let $\mathcal{F}$ be a w-partite family that satisfies the spread condition. We first define the function h. Since $\mathcal{F}$ is a w-partite family, there exists a partition of X to $X_1, \ldots, X_w$ such that for each $U \in \mathcal{F}$ and $j \in [w]$, $|X_j \cap U| = 1$.

Define the sample space as $X_1 \times \cdots \times X_w$. With a slight abuse of notation, we identify $\mathcal{F} \subseteq \mathcal{P}(X_1 \times \cdots \times X_w)$, and let D be a uniform distribution over $\mathcal{F}$. Since $\mathcal{F}$ is u^c-spread, the distribution D is a $(w, \log X, k)$ block min-entropy source with $k = c \log u \geq c \log w$. Then by Theorem 1.9, there exists seeds $r_1, \ldots, r_w$ such that $(\mathrm{IP}(D_1, r_1), \ldots, \mathrm{IP}(D_w, r_w))$ has full support, where $D = (D_1, \ldots, D_w)$. Note that the output of $\mathrm{IP}(\cdot, \cdot)$ is in $\{0,1\}^{k/c} \cong [u]$. We can now define h as follows:

$$h(x) = (\mathrm{IP}(x, r_j), j) \qquad \forall x \in X_j.$$

Note that by definition, h is injective on any $U \in X_1 \times \cdots \times X_w$. We identify elements of $\mathcal{U}^*$ with $\{(a_1, 1), \ldots, (a_w, w)\}$ with $a_i \in [u]$. Thus $h(\mathcal{F}) = \mathcal{U}^*$. The lemma now follows from Lemma 4.4 and Claim 4.3. □

We will also need the following lemma.

Lemma 4.5. Let $u \geq w$. Let c be the constant from Theorem 1.9. Then for every w-partite family $\mathcal{F}$ which is u^c-spread (recall Definition 1.13), it holds that $\mathcal{F}$ contains u pairwise disjoint sets.

Proof. The proof is very similar to the proof of Lemma 4.2. There is a map h for which $h(\mathcal{F}) = \mathcal{U}^*$. Note that $\mathcal{U}^*$ contains u pairwise disjoint sets, $U'_1, \ldots, U'_u$. By definition, $U'_i = h(U_i)$. But then also $U_1, \ldots, U_u$ must be pairwise disjoint. □

4.1 Sunflowers and robust sunflowers from compression

Now we can prove Theorem 1.8.

Theorem 4.6 (Theorem 1.8 restated). Suppose that there exists a strong $(k, 0, d, s)$-block min-entropy disperser, $E : \{0,1\}^n \times \{0,1\}^s \to \{0,1\}^d$ for any (w, n, k)-block min-entropy source. Then the following holds.

Let $\mathcal{F}$ be a w-uniform family. Assume that $|\mathcal{F}| \geq 2^{(k+2)w}$. Then:

(i) $\mathcal{F}$ contains a 2^d-sunflower.

(ii) $\mathcal{F}$ contains a $\left(p, w(1-p)^{2^d}\right)$-robust sunflower.

Proof. By Lemma 4.1, there is a w-partite subfamily $\mathcal{F}' \subseteq \mathcal{F}$ of size $|\mathcal{F}'| \geq 2^{kw}$. There are two possible cases.
Case 1: There is a subset $S \subseteq X$ such that

$$|\{U \in \mathcal{F}' : S \subseteq U\}| \geq |\mathcal{F}'| \cdot 2^{-k|S|}.$$

Define the family $\mathcal{F}'_S := \{U \setminus S : S \subseteq U \in \mathcal{F}'\}$. Notice that

- $\mathcal{F}'_S$ is $(w - |S|)$-partite;
- $|\mathcal{F}'_S| \geq |\mathcal{F}'| \cdot 2^{-k|S|} \geq 2^{k(w-|S|)}$.

By induction both (i) and (ii) hold.
Case 2: For all $S \subseteq X$,

$$|\{U \in \mathcal{F}' : S \subseteq U\}| \leq |\mathcal{F}'| \cdot 2^{-k|S|}$$

Notice that this is the spread condition for Lemma 4.2 and Lemma 4.5. Their conclusions are precisely (i) and (ii). □

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